

PA-M5-NL3-SV screened-shell candidate certificate and early causal no-go

TECT verification-first repository
claim T0-NON-CLAIM-PRE-A-CANDIDATE

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1. Purpose and scope: decision boundary

This note tests the new-hypothesis candidate PA-M5-NL3-SV-v0 introduced in EXP-000687. It is not a registered TECT action and inherits no A1, A2, A7, or A13 theorem. The calculation has a two-sided conclusion:

The candidate is retained as a coherent static screened-shell mathematical mechanism, but its bare form is rejected as a joint T-053 Lorentz-gauge-critical survivor.

The rejection is scoped to an isotropic nonzero soft shell with an ordinary positive inertial completion. It does not refute M5, nonlocal models, or emergent structure in general. A genuine gauge connection, isolated symmetry-protected nodes, or a separate light sector defines a materially new candidate and must rerun the reference-state theorem below.

On the cubic torus $\mathbb{T}_L^3$, let

$$\Psi : \mathbb{T}_L^3 \longrightarrow \mathbb{C}^3, \quad A_i : \mathbb{T}_L^3 \longrightarrow \mathbb{C}^3, \quad i = 1, 2, 3, \quad \rho = \Psi^\dagger \Psi. \quad (1.1)$$

For a Hermitian 3 by 3 matrix M , declare

$$\begin{aligned} \mathcal{F}[\Psi, A] = & \frac{1}{2} \langle \Psi, (rI + M - c\Delta)\Psi \rangle_{\mathbb{R}} + \frac{1}{2} \sum_i \langle A_i, (-\Delta + \sigma)A_i \rangle_{\mathbb{R}} \\ & - \sqrt{g} \sum_i \langle A_i, \partial_i \Psi \rangle_{\mathbb{R}} + \int_{\mathbb{T}_L^3} \left(\frac{u}{4} \rho^2 + \frac{v}{6} \rho^3 \right) dx. \end{aligned} \quad (1.2)$$

The parameter domain is

$$c > 0, \quad \sigma > 0, \quad g > 0, \quad v > 0, \quad r, u \in \mathbb{R}. \quad (1.3)$$

Equation (1.2), its boundary conditions, and its ordinary inertial extension are hypotheses. No numerical values, microscopic derivation, cooling map, fit, or measured-observable map is supplied.

2. Exact auxiliary elimination and coercivity

Put $L_\sigma = -\Delta + \sigma$. Since $\sigma > 0$, it is invertible and

$$A_i^* = \sqrt{g} L_\sigma^{-1} \partial_i \Psi. \quad (2.1)$$

Completing the square with the repository real pairing gives

$$\begin{aligned} \mathcal{F}[\Psi, A] &= \frac{1}{2} \sum_i \left\| L_\sigma^{1/2} (A_i - A_i^*) \right\|_2^2 + \frac{1}{2} \left\langle \Psi, \left(r\mathbf{I} + M - c\Delta - g(-\Delta)L_\sigma^{-1} \right) \Psi \right\rangle_{\mathbb{R}} \\ &\quad + \int_{\mathbb{T}_L^3} \left(\frac{u}{4} \rho^2 + \frac{v}{6} \rho^3 \right) dx. \end{aligned} \quad (2.2)$$

Thus, if $Mq_j = \lambda_j q_j$ and $s = |k|^2$, the reduced quadratic symbol is

$$\boxed{K_j(s) = r + \lambda_j + cs - \frac{gs}{s + \sigma}}. \quad (2.3)$$

There is no missing factor two: both quadratic terms in (1.2) carry one half, and the cross term carries exactly $-\sqrt{g}$.

Let $\lambda_{\min}$ be the lowest eigenvalue of M and set $\alpha = r + \lambda_{\min} - g$. Since

$$K_j(s) \geq cs + r + \lambda_j - g, \quad (2.4)$$

and the scalar polynomial

$$\frac{\alpha x}{2} + \frac{ux^2}{4} + \frac{vx^3}{12}$$

has a finite lower bound on $x \geq 0$, (2.2) yields

$$\mathcal{F} \geq \frac{1}{2} \sum_i \left\| L_\sigma^{1/2} (A_i - A_i^*) \right\|_2^2 + \frac{c}{2} \|\nabla \Psi\|_2^2 + \frac{v}{12} \|\Psi\|_6^6 - |\mathbb{T}_L^3| C_{\text{pol}}. \quad (2.5)$$

Moreover

$$\sum_i \|L_\sigma^{-1} \partial_i \Psi\|_{H^1}^2 \leq \sup_{s \geq 0} \frac{(1+s)s}{(s+\sigma)^2} \|\Psi\|_2^2, \quad \sup_{s \geq 0} \frac{(1+s)s}{(s+\sigma)^2} \leq \max\{1, \sigma^{-1}\}. \quad (2.6)$$

Therefore finite-torus sublevel sets are bounded in $H^1(\mathbb{T}_L^3; \mathbb{C}^3) \times H^1(\mathbb{T}_L^3; \mathbb{C}^9)$. Weak compactness, compact $H^1 \hookrightarrow L^4$, and lower semicontinuity of the positive quadratic and sextic terms give a global static minimizer. This is a static well-posedness statement, not a dynamical or causal theorem.

3. Continuous and finite-torus shell gates are different

Direct differentiation of (2.3) gives

$$K'_j(s) = c - \frac{g\sigma}{(s+\sigma)^2}, \quad K''_j(s) = \frac{2g\sigma}{(s+\sigma)^3} > 0. \quad (3.1)$$

Hence the continuous radial minimum is nonzero exactly when

$$g > c\sigma, \quad s_* = \sqrt{g\sigma/c} - \sigma > 0, \quad (3.2)$$

and then

$$K_j(s_*) = r + \lambda_j - (\sqrt{g} - \sqrt{c\sigma})^2, \quad (3.3)$$

with the useful exact identity

$$K_j(s) - K_j(s_*) = \frac{c(s - s_*)^2}{s + \sigma}. \quad (3.4)$$

Condition (3.2) selects a radial minimum; it does not assert that this mode is soft or unstable. Those cases require respectively $K_j(s_*) = 0$ or $K_j(s_*) < 0$.

On the finite cubic torus,

$$s_n = \left(\frac{2\pi}{L}\right)^2 |n|^2, \quad s_1 = \left(\frac{2\pi}{L}\right)^2. \quad (3.5)$$

Since

$$K_j(s) - K_j(0) = s \left(c - \frac{g}{s + \sigma} \right), \quad (3.6)$$

some nonzero lattice mode lies strictly below zero momentum if and only if

$$\boxed{g > c(\sigma + s_1)}. \quad (3.7)$$

At equality the zero mode and first shell are degenerate. The exact counterfixture $c = \sigma = 1$, $g = 3/2$, $L = 2\pi$ satisfies the continuous condition (3.2), but every nonzero torus mode has $s \geq 1$ and lies above the zero mode. This corrects any use of $g > c\sigma$ as a finite-volume shell selection theorem.

4. Exact neutral-reference theorem and morphology

Let

$$\kappa_L = \min_{n \in \mathbb{Z}^3, j} K_j(s_n), \quad u_- := \max\{-u, 0\}, \quad u_+ := \max\{u, 0\}. \quad (4.1)$$

For $x \geq 0$, exact completion gives

$$\frac{\kappa_L x}{2} + \frac{ux^2}{4} + \frac{vx^3}{6} = x \left[\frac{1}{2} \left(\kappa_L - \frac{3u_-^2}{16v} \right) + \frac{u_+ x}{4} + \frac{v}{6} \left(x - \frac{3u_-}{4v} \right)^2 \right]. \quad (4.2)$$

The spectral lower bound and (4.2), together with a minimum-mode plane wave that saturates both, prove

$$\boxed{\Psi = 0 \text{ is a global minimizer} \iff \kappa_L \geq \frac{3u_-^2}{16v}}. \quad (4.3)$$

If the inequality is strict, zero is the unique global minimizer. If it is violated, coercivity supplies a nonzero global minimizer. When $u < 0$, equality gives first-order coexistence at

$$\rho_t = \frac{3u_-}{4v}, \quad \kappa_L = \frac{3u_-^2}{16v} > 0. \quad (4.4)$$

Thus the zero phase is still quadratically gapped at the actual coexistence boundary. The massless shell at $\kappa_L = 0$ is a later spinodal, not the first-order transition. If $u \geq 0$, the continuous onset can instead occur at $\kappa_L = 0$, but Section 5 then fires.

The inherited A1 family-lock matrix is

$$M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}. \quad (4.5)$$

Its characteristic polynomial and discriminant are

$$t^3 - \frac{2}{5}t^2 + \frac{223}{5000}t - \frac{3}{3125}, \quad \frac{32233}{31250000000} > 0, \quad (4.6)$$

so the lowest eigendirection is simple. In the nonzero-minimum regime, equality in (2.2) and (4.2) forces $A = A^*$, Fourier support on the minimum shell in that eigendirection, and constant $|\Psi|$. The remaining finite Fourier sum is a Laurent polynomial P satisfying $P(z)P^*(z) = \rho_t$. It is therefore a unit of

$\mathbb{C}[z_1^{\pm 1}, z_2^{\pm 1}, z_3^{\pm 1}]$, whose only units are az^n . Every nonzero global minimizer is consequently a single plane wave

$$\Psi(x) = \sqrt{\rho_*} q_{\min} e^{ik_{\min} \cdot x}, \quad (4.7)$$

up to phase and the discrete shell degeneracy. This is a uniform-density phase twist with an unselected direction, not a density lattice or BCC state. The finite-volume level crossing in (4.3) is not yet a thermodynamic or statistical phase transition; robustness in L and an infinite-volume Gibbs analysis remain open.

5. Ordinary inertial completion and the causal no-go

A causal question exists only after separately hypothesising positive kinetic terms

$$\frac{\chi_\Psi}{2} \|\dot{\Psi}\|_2^2 + \frac{\chi_A}{2} \sum_i \|\dot{A}_i\|_2^2, \quad \chi_\Psi, \chi_A > 0. \quad (5.1)$$

For the longitudinal A component, the exact characteristic equation is

$$(D_j(s) - \chi_\Psi \omega^2)(L(s) - \chi_A \omega^2) - gs = 0, \quad (5.2)$$

where $D_j = r + \lambda_j + cs$ and $L = s + \sigma$. The two transverse auxiliary-vector branches satisfy

$$\omega_T^2 = \frac{s + \sigma}{\chi_A}. \quad (5.3)$$

At high momentum, the two principal speeds are

$$\sqrt{c/\chi_\Psi}, \quad 1/\sqrt{\chi_A}. \quad (5.4)$$

One ultraviolet cone therefore requires the unprotected tuning

$$c\chi_A = \chi_\Psi. \quad (5.5)$$

More decisively, tune one continuous shell critical so that $K_{j_*}(s_*) = 0$ and write $k = k_* + p$, $|k_*|^2 = s_*$, $p = p_{\parallel} \hat{k}_* + p_{\perp}$. With

$$B_* := \chi_A D_{j_*}(s_*) + \chi_\Psi(s_* + \sigma) > 0,$$

equations (3.4) and (5.2) give

$$\boxed{\omega_-^2(k_* + p) = \frac{c}{B_*} \left(2\sqrt{s_*} p_{\parallel} + p_{\parallel}^2 + |p_{\perp}|^2 \right)^2 + O(|s - s_*|^3)}. \quad (5.6)$$

In particular,

$$\omega_- \asymp |p_{\parallel}| \quad \text{radially}, \quad \omega_- \asymp |p_{\perp}|^2 \quad \text{tangentially}. \quad (5.7)$$

The spatial quadratic Hessian has rank one, not rank three. Equation (5.6) is a curved-shell square: paths that remain on $|k|^2 = s_*$ stay exactly soft. It must not be replaced by a false global comparison with $p_{\parallel}^2 + |p_{\perp}|^4$.

No choice of c, g, σ and no high-frequency tuning (5.5) creates the missing tangential $p_{\perp}^2$ term. Therefore a rotationally invariant nonzero soft sphere with regular positive inertia cannot supply a positive isotropic Lorentz cone. When $u < 0$, the actual first-order boundary is gapped by (4.4); when $u \geq 0$, the massless boundary has the rank-one shell defect. The bare candidate fails in both cases.

Also, $\sigma > 0$ makes (5.3) gapped at zero momentum. Taking $\sigma = 0$ removes invertibility and coercive control of the A zero mode, while the nonzero shell collapses toward zero momentum. The same bare A_i cannot both generate the screened finite shell and act as a massless photon.

6. Gauge, symmetry, and observable audit

The fields $A_i \in \mathbb{C}^3$ transform as vector matter, not as a gauge connection. Under a local common phase multiplication, derivatives of the phase enter both $\partial_i \Psi$ and ∇A_i ; (1.2) is not locally $U(1)$ invariant. It contains no A_0 , Gauss constraint, inhomogeneous connection transformation, Maxwell curvature, or gauge quotient.

If A remains auxiliary, (2.1) is an instantaneous Yukawa-type spatial response and supplies no microscopic finite-propagation theorem. If (5.1) is added, A is no longer auxiliary and contributes extra propagating modes and generally a second cone. Either interpretation blocks identifying this screened vector with a photon.

Because (4.6) is simple, the unitary commutant of literal M is generically $U(1)^3$, not $U(3)$. No $\mathbb{CP}^2$ quotient follows. A global phase circle can support a limited winding discussion inside a nonzero-amplitude sector, but amplitude zeros can unwind it, and no flux quantum, charge, $\hbar$, or e is derived.

For (4.7), ρ , internal bilinears, and $|A|^2$ are uniform. The phase gradient and tensors such as $\partial_i \Psi^\dagger \partial_j \Psi = \rho k_i k_j$ expose a preferred direction. Under a future local gauge completion that phase gradient may instead be gauge equivalent to a uniform state. In neither interpretation does the present candidate supply a derived gauge-invariant density crystal or measured-observable map.

7. Devil's-advocate audit

1. **Objection:** $g > c\sigma$ already proves a finite-torus shell. **UPHELD as false.** The exact discrete condition is (3.7), and the $g = 3/2$ counterfixture separates the two claims.
2. **Objection:** the first-order level crossing supplies the desired massless critical mode. **UPHELD as false for $u < 0$.** Equation (4.4) shows a strictly positive quadratic gap at coexistence.
3. **Objection:** tuning the two ultraviolet speeds repairs the shell. **UPHELD as false.** Relation (5.5) leaves the rank-one Hessian and the tangential quadratic-frequency law unchanged.
4. **Objection:** nonlinear crystal selection could isolate directions and restore an ordinary cone. **VALID, but it defines an unproved new candidate.** The exact bare minimizer is instead the plane-wave family (4.7). Angular operators, additional fields, or a distinct interaction must be declared before testing and must rerun boundedness and the zero-reference gate.
5. **Objection:** the screened vector is already a gauge field. **UPHELD as false.** Its transformation law, field content, constraint structure, transverse gap, and principal symbol do not define a gauge connection.
6. **Objection:** the inherited M provides $U(3)$ or $\mathbb{CP}^2$. **DISMISSED by (4.6).** Its simple spectrum leaves only the commuting phase rotations; no local quotient is present.
7. **Objection:** finite-volume minimization is already a thermodynamic phase transition. **UPHELD as false.** No $L \rightarrow \infty$ free-energy nonanalyticity, Gibbs measure, robustness envelope, or cooling dynamics is proved.
8. **Objection:** exact algebra may hide a sign, factor, convention, or hardcoded answer. **VALID WITH MITIGATION.** The primary SymPy certificate and a non-importing standard-library Fraction implementation independently reconstruct the A1 matrix, factor-one-half convention, auxiliary completion, continuum and discrete shell gates, reference-state threshold, coercive factors, dynamic rank, and adversarial fixtures. External review is invited for units, Fourier normalization, the direct-method passage, the Laurent-unit argument, the inertial hypothesis, and all scope firewalls.

8. Reproduction and Result footer

Run from the repository root:

```
$py = "E:\Dev\TECT.venv\Scripts\python.exe"  
& $py codes/foundations/pre_a_pa_m5_n13_sv_candidate.py  
& $py codes/foundations/pre_a_pa_m5_n13_sv_candidate_independent.py  
& $py codes/foundations/pre_a_pa_m5_n13_sv_candidate_verify.py
```

The primary certificate must report 41/41 PASS. The independent certificate uses only Python's standard library and must report 29/29 PASS. The integrated verifier reruns both in temporary locations and audits parity, hashes, the candidate manifest, this note and PDF, the negative and exploration records, T-054 routing, generated maps, and the release regressions.

Candidate ID: PA-M5-NL3-SV-v0.

Task: T-054.

Authority: T0 candidate certificate; claim_bearing=false; no result-ledger ID.

Exact static result: auxiliary elimination gives

$K_j(s)=r+\lambda_j+c*s-g*s/(s+\sigma)$; finite-torus zero selection is
equivalent to $\kappa_L>3*u_{\text{minus}}^2/(16*v)$.

Formal negative: NG-2026-08-03-PA-M5-BARE-ISOTROPIC-SHELL-CAUSAL-CONE.

Negative statement: at a nonzero isotropic critical shell and regular positive inertia, the spatial quadratic Hessian has rank one and the tangential frequency is quadratic. For $u<0$ the first-order boundary is instead gapped.

Candidate verdict: RETAIN STATIC SCREENED-SHELL MATHEMATICAL MECHANISM;
REJECT BARE JOINT T-053 LORENTZ-GAUGE-CRITICAL SURVIVOR.

Scope: finite periodic static functional (1.2), $c,\sigma,g,v>0$, Hermitian M ;
causal test only for the separately declared ordinary positive inertia (5.1).

No-overclaim: no microscopic derivation, parameter fit, thermodynamic limit,
gauge completion, photon, CP2, physical vacuum, BCC, density lattice,
topology, A7, T-050/A13, Sector-A closure, or TECT tier action follows.

Next action: preserve this static lemma and move the tournament to a genuine
compact-gauge M3 candidate or an M5 candidate with protected isolated nodes.

Decision footer.